INCH-POUND

MIL-G-47035A (MI)
27 September 1991
SUPERSEDING
MIL-G-47035 (MI)
19 April 1974

MILITARY SPECIFICATION

GLASS, BARIUM, SODA, POTASH

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of potash, soda, barium, glass.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9340

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SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-D-723	-	Drums, Fiber
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 through 12 Gallon)

MILITARY

MIL-C-3955	-	Cans, Composite, Spirally Wound

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

ARMY

10151439	-	Bead, Glass

(Copies of the drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified,

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the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 336	-	Annealing Point and Strain
ASTM C 338	-	Softening Point of Glass, Standard Test Method for
ASTM D 792	-	Specific Gravity (Relative Density) and Density of Plastics by Displacement, Standard Test Methods for
ASTM E 111	-	Young's Modulus, Tangent Modulus, and Chord Modulus, Standard Test Method for
ASTM E 228	-	Linear Thermal Expansion of Rigid Solid Materials With Vitreous Silica Dilatometer, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Material. The material shall be an annealed glass formulated primarily from silica, potassium oxide, sodium oxide, and barium oxide.

3.3 Mechanical properties.

3.3.1 Coefficient of expansion. The coefficient of thermal expansion shall be 82.0 $\pm$ 2.5 times 10^{-7} inches per inch per degree Celsius (C) in the range of 0 to 300°C (see 4.7.2.1).

3.3.2 Modulus of elasticity. The modulus of elasticity shall be not less than 10.2 times 10^{6} pounds force per square inch (lbf/in^2) (see 4.7.2.2).

3.4 Physical properties.

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3.4.1 Specific gravity. The specific gravity shall be 2.610 ± 0.005 (see 4.7.3.1).

3.4.2 Annealing point. The annealing point shall be 475 ± 10°C (see 4.7.3.2).

3.4.3 Softening point. The softening point shall be 645 ± 5°C (see 4.7.3.3).

3.4.4 Strain point. The strain point shall be 432 ± 5°C (see 4.7.3.4).

3.4.5 Dimensions. The glass shall be in the form of beads with dimensions as specified on Drawing 10151439.

3.5 Workmanship. The workmanship shall be such as to ensure a high quality product which is uniform, and in conformance with this specification. The glass shall be free of dirt, foreign materials, cracks, or contaminants.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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4.3 First article inspection. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with all the tests specified herein.

4.3.1 First article sample. The first article sample shall be prepared using the same methods proposed for the preparation of subsequent lots of material. First article samples which do not meet the requirements of this specification shall be rejected and returned to the supplier. Subsequent quantities will not be considered for acceptance until approval of the first article sample has been obtained.

4.3.2 First article tests. First article tests shall be conducted only on the first article sample and shall consist of all the examinations and tests specified herein.

4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in table I.

TABLE I. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Material	3.2	4.6
Coefficient of expansion	3.3.1	4.7.2.1
Modulus of elasticity	3.3.2	4.7.2.2
Specific gravity	3.4.1	4.7.3.1
Annealing point	3.4.2	4.7.3.2
Softening point	3.4.3	4.7.3.3
Strain point	3.4.4	4.7.3.4
Dimensions	3.4.5	4.7.3.5

4.5 Lot size. A lot shall consist of all glass beads manufactured at one time from the same batch of glass, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.6 Sampling. Unless otherwise specified (see 6.2), ten glass beads from one unit, representative of each lot shall be selected at random for quality conformance testing. Each container in a lot shall be considered a unit of product. Failure of one or more beads in the sample to meet quality conformance test requirements shall be cause for lot rejection.

4.7 Test methods.

4.7.1 Test conditions. Unless otherwise specified in the applicable test, all tests shall be conducted at 23 $\pm$ 2°C at a relative humidity of 50 $\pm$ 5 percent (%).

4.7.2 Mechanical properties tests.

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4.7.2.1 Coefficient of expansion. The coefficient of expansion shall be determined in accordance with ASTM E 228. The coefficient of expansion shall be as specified in 3.3.1.

4.7.2.2 Modulus of elasticity. The modulus of elasticity shall be determined in accordance with ASTM E 111 and shall be as specified in 3.3.2

4.7.3 Physical properties tests.

4.7.3.1 Specific gravity. The specific gravity shall be determined in accordance with ASTM D 792, Method A and shall be as specified in 3.4.1.

4.7.3.2 Annealing point. The specific gravity shall be determined in accordance with ASTM C 336, and shall be as specified in 3.4.2.

4.7.3.3 Softening point. The softening point shall be determined in accordance with ASTM C 338, and shall be as specified in 3.4.3.

4.7.3.4 Strain point. The strain point shall be determined in accordance with ASTM C 336, and shall be as specified in 3.4.4.

4.7.3.5 Dimensions. The dimensions of the glass beads shall be measured with an instrument capable of measuring to the tolerance specified. The dimensions shall be as specified in 3.4.5.

4.8 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with MIL-STD-1190.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.1.1 Level A. Unless otherwise specified in the contract or purchase order, the glass beads shall be packaged in metal containers conforming to PPP-C-96, or PPP-P-704; fiber containers conforming to PPP-B-636, PPP-D-723, or MIL-C-3955; or equivalent plastic containers. Size of the container shall be as specified by the procuring activity. Containers shall be non-reactive to the contents.

5.1.2 Level B. Same as Level A.

5.1.3 Level C. The material shall be packaged in accordance with MIL-STD-1190.

5.2 Packing.

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5.2.1 Level A. The material packaged in accordance with 5.1 shall be packed in overseas type containers conforming to PPP-B-601, PPP-B-621, PPP-B-636, or equivalent containers suitable for the intended purpose.

5.2.2 Level B. Same as Level A.

5.2.3 Level C. Materials packaged as specified in 5.1.3 shall be packed in accordance with MIL-STD-1190.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The glass is intended as an electrical insulating seal.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Whether a first article is required and if so, pertinent details (see 3.1)

d. Conditions for quality conformance inspection (see 4.4)

e. Sampling plan if other than specified (see 4.6)

f. Requirements for preservation, packaging, packing, and marking if different (see 5.1)

g. Size of container (see 5.1.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article

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inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Insulator
Electrical suppressor
Frit

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Review activity:
DLA-GS

Preparing Activity:
Army - MI

(Project No. 9340-A073)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-G-47035 (MI)	910927

3. DOCUMENT TITLE

GLASS, BARIUM, SODA, POTASH

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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